

Jurnal Praktikum

Sistem Operasi

Modul 15: Keamanan

Catatan

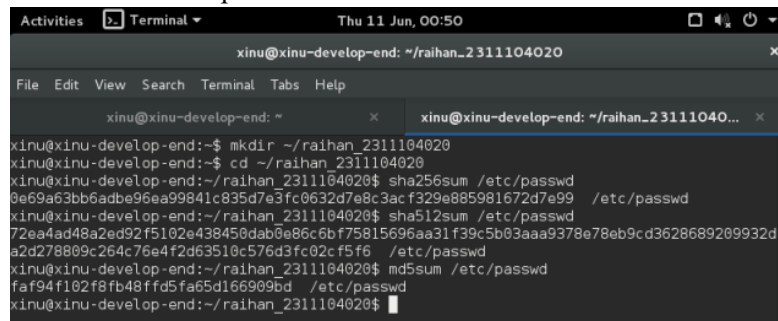
1. Praktikan wajib untuk screenshot setiap langkah yang dikerjakan hingga tampilan output akhir
2. Untuk soal source code, kumpulkan SS-nya saja
3. Berikan identitas nama - nim dalam bentuk comment di Source Code

JANGAN LUPA UNTUK HAPUS FILE DAN SCREENSHOT DI UBUNTU DAN WINDOWS

Buatlah folder di home dengan nama "Nama_NIM", simpan semua file jurnal yang Anda kerjakan di folder tersebut

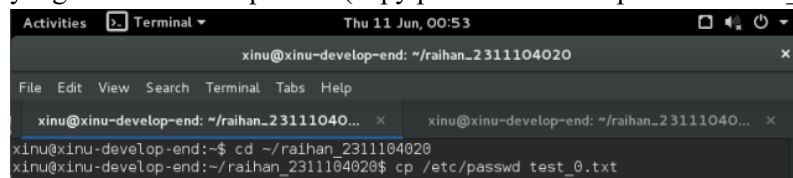
1. Integritas: dasar hashing [10 Point]

- a. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file /etc/passwd. Berapa nilai hash dari file /etc/passwd? Screenshot nilai hash dari file tersebut.



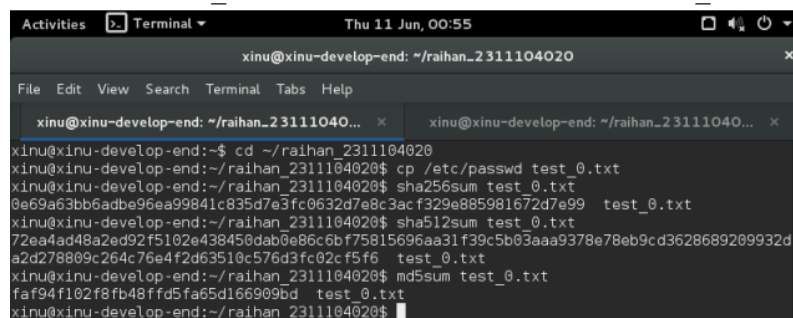
```
xinu@xinu-develop-end: ~/$ mkdir ~/raihan_2311104020
xinu@xinu-develop-end: ~/$ cd ~/raihan_2311104020
xinu@xinu-develop-end: ~/raihan_2311104020$ sha256sum /etc/passwd
0e69a63bb6adbe96ea99841c835d7e3fc0632d7e8c3acf329e885981672d7e99 /etc/passwd
xinu@xinu-develop-end: ~/raihan_2311104020$ sha512sum /etc/passwd
72ea4ad48a2ed92f5102e438450dab0e86c6bf75815696aa31f39c5b03aaa9378e78eb9cd3628689209932d
a2d278809c264c76e4f2d63510c576d3fc02cf5f6 /etc/passwd
xinu@xinu-develop-end: ~/raihan_2311104020$ md5sum /etc/passwd
faf94f102f8fb48ffd5fa65d166909bd /etc/passwd
xinu@xinu-develop-end: ~/raihan_2311104020$
```

- b. [2 Point] Buatlah file bernama test_0.txt pada folder /home/praktikan. Isi file tersebut isi yang ada di file /etc/passwd (copy paste isi file /etc/passwd ke test_0.txt)



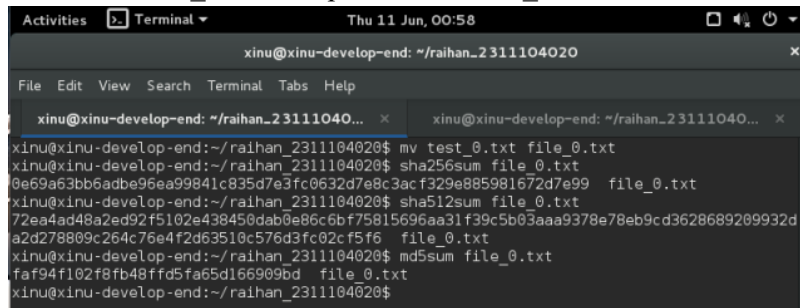
```
xinu@xinu-develop-end: ~/$ cd ~/raihan_2311104020
xinu@xinu-develop-end: ~/raihan_2311104020$ cp /etc/passwd test_0.txt
```

- c. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_0.txt. Berapa nilai hash dari file test_0.txt? Screenshot nilai hash dari file test_0.txt.



```
xinu@xinu-develop-end: ~/$ cd ~/raihan_2311104020
xinu@xinu-develop-end: ~/raihan_2311104020$ cp /etc/passwd test_0.txt
xinu@xinu-develop-end: ~/raihan_2311104020$ sha256sum test_0.txt
0e69a63bb6adbe96ea99841c835d7e3fc0632d7e8c3acf329e885981672d7e99 test_0.txt
xinu@xinu-develop-end: ~/raihan_2311104020$ sha512sum test_0.txt
72ea4ad48a2ed92f5102e438450dab0e86c6bf75815696aa31f39c5b03aaa9378e78eb9cd3628689209932d
a2d278809c264c76e4f2d63510c576d3fc02cf5f6 test_0.txt
xinu@xinu-develop-end: ~/raihan_2311104020$ md5sum test_0.txt
faf94f102f8fb48ffd5fa65d166909bd test_0.txt
xinu@xinu-develop-end: ~/raihan_2311104020$
```

- d. [2 Point] Rename file test_0.txt menjadi file_0.txt. Lakukan hash SHA256, SHA512 dan MD5 untuk file_0.txt. Berapa nilai hash file_0.txt? Screenshot nilai hash dari file_0.txt.



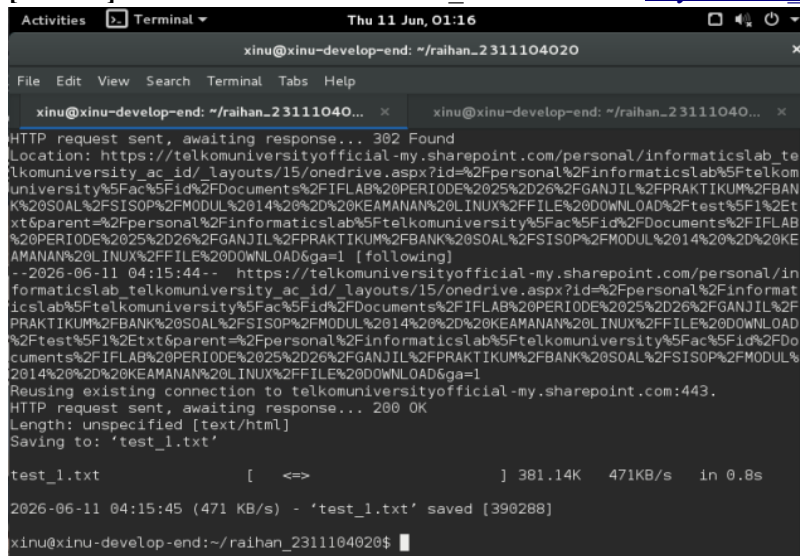
```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/raihan_2311104020
xinu@xinu-develop-end:~/raihan_2311104020$ mv test_0.txt file_0.txt
xinu@xinu-develop-end:~/raihan_2311104020$ sha256sum file_0.txt
0e69a63bb6adbe96ea99841c835d7e3fc0632d7e8c3acf329e885981672d7e99  file_0.txt
xinu@xinu-develop-end:~/raihan_2311104020$ sha512sum file_0.txt
72ea4ad48a2ed92f5102e438450dab0e86c6bf75815696aa31f39c5b03aaa9378e78eb9cd3628689209932d
a2d278809c264c76e4f2d63510c576d3fc02cf5f6  file_0.txt
xinu@xinu-develop-end:~/raihan_2311104020$ md5sum file_0.txt
faf94f102f8fb48ffd5fa65d166909bd  file_0.txt
xinu@xinu-develop-end:~/raihan_2311104020$
```

- e. [2 Point] Apa hasil pengamatan Anda? File apa saja yang mempunyai hash yang sama? Jelaskan!

File /etc/passwd, text_0.txt, dan file_0.txt memiliki nilai hash yang sama persis. Hal ini terjadi karena algoritma hashing hanya mengevaluasi konten (isi data) dari sebuah file. Meskipun nama file diubah atau disalin, selama isi datanya tidak diubah sedikit pun, nilai hash akan tetap identik.

2. Integritas: avalance [15 Point]

- a. [3 Point] Download file bernama test_1.txt di link ini tiny.cc/test1_txt



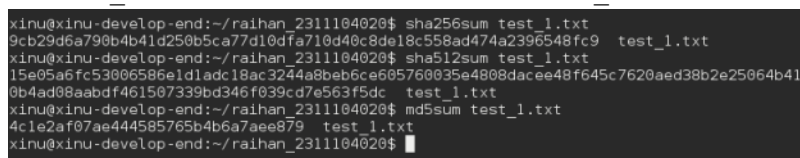
```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/raihan_2311104020
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finformaticslab%2Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20S0AL%2FSISOP%2FMODULE%2014%20%2D20KEAMANAN%20LINUX%2FFILE%20DOWNL0AD%2Ftest%5F1%2Etxt&parent=%2Fpersonal%2Finformaticslab%2Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20S0AL%2FSISOP%2FMODULE%2014%20%2D20KEAMANAN%20LINUX%2FFILE%20DOWNL0AD&ga=1 [following]
--2026-06-11 04:15:44-- https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finformaticslab%2Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20S0AL%2FSISOP%2FMODULE%2014%20%2D20KEAMANAN%20LINUX%2FFILE%20DOWNL0AD&ga=1
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test_1.txt'

test_1.txt          [  <=>          ] 381.14K  471KB/s  in 0.8s

2026-06-11 04:15:45 (471 KB/s) - 'test_1.txt' saved [390288]

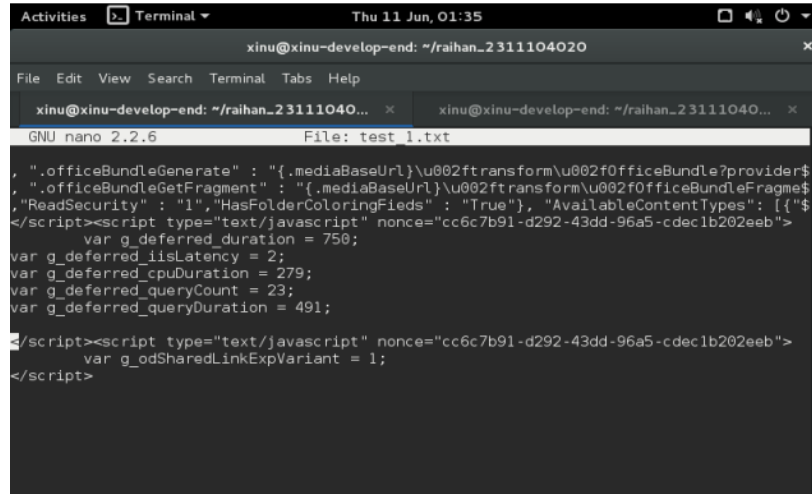
xinu@xinu-develop-end:~/raihan_2311104020$
xinu@xinu-develop-end:~/raihan_2311104020$ sha256sum test_1.txt
9cb29d6a798b4b41d250b5ca77d10dfa710d40c8de18c558ad474a2396548fc9  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$ sha512sum test_1.txt
15e05a6fc53806586e1d1adc18ac3244a8beb6ce605760035e4808dacee48f645c7620aed38b2e25064b410
0b4ad08aabd461507339bd346f039cd7e563f5dc  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$ md5sum test_1.txt
4c1e2af07ae444585765b4b6a7aee879  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$
```

- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.txt. Berapa nilai hash test_1.txt? Screenshot nilai hash dari file test_1.txt.

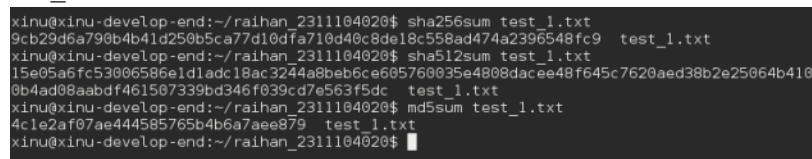


```
xinu@xinu-develop-end:~/raihan_2311104020$ sha256sum test_1.txt
9cb29d6a798b4b41d250b5ca77d10dfa710d40c8de18c558ad474a2396548fc9  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$ sha512sum test_1.txt
15e05a6fc53806586e1d1adc18ac3244a8beb6ce605760035e4808dacee48f645c7620aed38b2e25064b410
0b4ad08aabd461507339bd346f039cd7e563f5dc  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$ md5sum test_1.txt
4c1e2af07ae444585765b4b6a7aee879  test_1.txt
xinu@xinu-develop-end:~/raihan_2311104020$
```

- c. [3 Point] Hapuslah titik diakhir file test_1.txt tersebut, simpan file tersebut!



- d. [3 Point] Lakukan hash dari SHA256, SHA512 dan MD5. Screenshot nilai hash dari file test_1.txt.

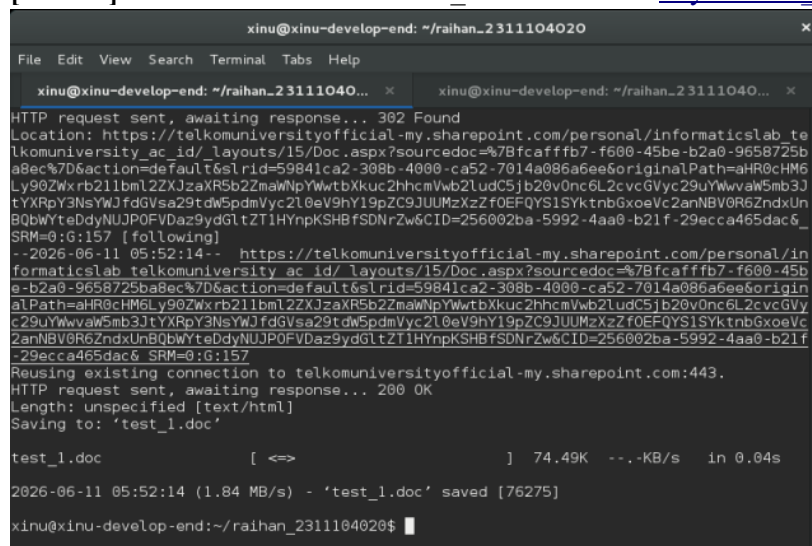


- e. [3 Point] Apa analisis (hasil pengamatan) Anda mengenai hal tersebut! Apakah nilai hash sama?

Nilai hash berubah secara drastic dan sama sekali tidak mirip dengan hash awal meskipun hanya satu karakter (titik) yang dihapus. Fenomena ini disebut *Avalanche Effect*, di mana perubahan sekecil apa pun pada data input akan menghasilkan nilai hash yang berubah total.

3. Integritas: metadata [18 Point]

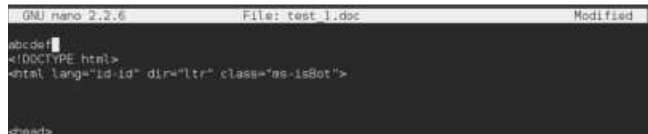
- a. [3 Point] Download file bernama test_1.doc di link ini tiny.cc/test1_doc



- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.

```
xinu@xinu-develop-end:~/raihan_2311104020$ sha256sum test_1.doc
98a2364a6abf4af7ef0661a700952b91d592563496f629d904e46d86bd1f57c7 test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$ sha512sum test_1.doc
0f04d46719a6859c5d9c1c32790da838b13068a4ec5d35f3a8a9f5713f882dc2a8d257dfaabb8ac247e2ec6
2bb9c603a6a8bb8d4d7faf1afffd266c9011cf55a test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$ md5sum test_1.doc
f1eb2cdc602e90c2decdd86a5d5cf947f test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$
```

- c. Buka kembali file test_1.doc. Lakukan hal ini:
- [3 Point]** Ketik abcdef. Save test_1.doc
 - [3 Point]** Hapus abcdef. Save test_1.doc



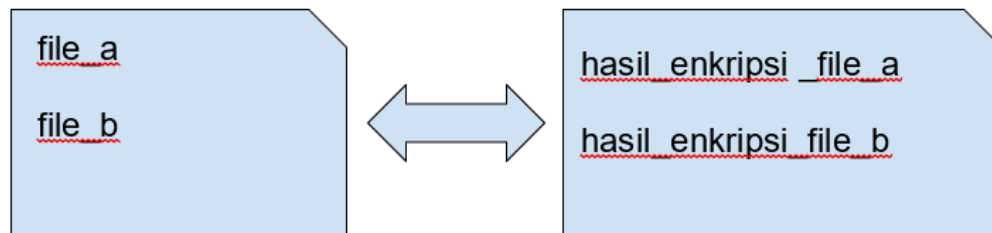
- d. **[3 Point]** Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.

```
xinu@xinu-develop-end:~/raihan_2311104020$ sha256sum test_1.doc
98a2364a6abf4af7ef0661a700952b91d592563496f629d904e46d86bd1f57c7 test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$ sha512sum test_1.doc
0f04d46719a6859c5d9c1c32790da838b13068a4ec5d35f3a8a9f5713f882dc2a8d257dfaabb8ac247e2ec6
2bb9c603a6a8bb8d4d7faf1afffd266c9011cf55a test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$ md5sum test_1.doc
f1eb2cdc602e90c2decdd86a5d5cf947f test_1.doc
xinu@xinu-develop-end:~/raihan_2311104020$
```

- e. **[3 Point]** Hasil pengamatan apa yang diperoleh? Jelaskan alasannya!
 Nilai hash menjadi berbeda dari file awalnya. Alasannya adalah file dengan format .doc menyimpan metadata (informasi tentang data), seperti waktu modifikasi terakhir (timestamp) dan jumlah revisi dokumen. Setiap kali kita melakukan fungsi save, metadata tersebut ikut diperbarui, sehingga meskipun isi teks dikembalikan seperti semula, struktur file keseluruhan telah berubah dan menghasilkan nilai hash yang berbeda.

4. **Konfidensialitas: encfs [27 Point]**

Encfs adalah tool untuk melakukan mount dan membuat file system yang terenkripsi. Cara kerjanya adalah sebagai berikut. Ada 2 folder yang akan dibuat, dua folder tersebut saling terkait. Satu folder merupakan folder yang tidak terenkripsi, folder lainnya merupakan hasil enkripsi dari folder pertama.



WARNING: INSTALL ENCFs!

```
sudo apt-get update
sudo apt-get -y install encfs
```

Setelah terinstall silakan melanjutkan pengerjaan jurnal.

- a. **[5 Point]** Jalankan perintah berikut:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_normal
```

Pilih y (enter), pilih y (enter), enter, kemudian buatlah password. **Ingatlah password yang dibuat.** Setelah selesai, perhatikan bahwa telah muncul folder bernama folder_normal dan folder_terenkripsi.

```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/... x xinu@xinu-develop-end: ~/... x xinu@xinu-develop-end: ~/... x
Preparing to unpack .../libboost-filesystem1.55.0_1.55.0+dfsg-3_i386.deb ...
Unpacking libboost-filesystem1.55.0:i386 (1.55.0+dfsg-3) ...
Selecting previously unselected package libboost-serialization1.55.0:i386.
Preparing to unpack .../libboost-serialization1.55.0_1.55.0+dfsg-3_i386.deb ...
Unpacking libboost-serialization1.55.0:i386 (1.55.0+dfsg-3) ...
Selecting previously unselected package librlog5.
Preparing to unpack .../librlog5_1.4-2.1_i386.deb ...
Unpacking librlog5 (1.4-2.1) ...
Selecting previously unselected package encfs.
Preparing to unpack .../encfs_1.7.4-5_i386.deb ...
Unpacking encfs (1.7.4-5) ...
Processing triggers for man-db (2.7.0.2-5) ...
Setting up libboost-filesystem1.55.0:i386 (1.55.0+dfsg-3) ...
Setting up libboost-serialization1.55.0:i386 (1.55.0+dfsg-3) ...
Setting up librlog5 (1.4-2.1) ...
Setting up encfs (1.7.4-5) ...
Processing triggers for libc-bin (2.19-18) ...
xinu@xinu-develop-end:~/raihan_2311104020$ encfs ~/raihan_2311104020/folder_terenkripsi
~/raihan_2311104020/folder_normal
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?> 
```

- b. [5 Point] Copy dua atau tiga buah file (file apa saja) ke folder_normal. Amati dan tulis hasil observasi Anda pada folder_terenkripsi!

```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/... x xinu@xinu-develop-end: ~/... x xinu@xinu-develop-end: ~/... x
xinu@xinu-develop-end:~/raihan_2311104020$ cp etc/passwd ~/raihan_2311104020/folder_normal
cp: cannot stat 'etc/passwd': No such file or directory
xinu@xinu-develop-end:~/raihan_2311104020$ 
xinu@xinu-develop-end:~/raihan_2311104020$ cp /etc/passwd ~/raihan_2311104020/folder_normal
cp: cannot create regular file '~/raihan_2311104020/folder_normal': No such file or directory
xinu@xinu-develop-end:~/raihan_2311104020$ cp /etc/passwd ~/raihan_2311104020/folder_normal/
cp: cannot create regular file '~/raihan_2311104020/folder_normal/': No such file or directory
xinu@xinu-develop-end:~/raihan_2311104020$ cp /etc/passwd folder_normal/
xinu@xinu-develop-end:~/raihan_2311104020$ ls folder_normal
passwd
xinu@xinu-develop-end:~/raihan_2311104020$ ls folder_terenkripsi
OAVUnUSNms3wVBiP,0XIjL,b
xinu@xinu-develop-end:~/raihan_2311104020$ 
```

- c. [5 Point] Hapus salah satu file (bebas) pada folder_terenkripsi. Amati dan tulis hasil observasi Anda pada folder_normal!

```
xinu@xinu-develop-end:~/raihan_2311104020$ ls folder_terenkripsi
OAVUnUSNms3wVBiP,0XIjL,b
xinu@xinu-develop-end:~/raihan_2311104020$ rm ~/raihan_2311104020/folder_terenkripsi/OAVUnUSNms3wVBiP,0XIjL,b
xinu@xinu-develop-end:~/raihan_2311104020$ ls folder_normal
xinu@xinu-develop-end:~/raihan_2311104020$ 
```

- d. [5 Point] Lakukan umount dengan perintah:

```
Fusermount -u ~/folder_anda/folder_normal
```

Amati dan tulis hasil observasi Anda pada folder_normal dan folder_terenkripsi

```
xinu@xinu-develop-end:~/raihan_2311104020$ ls folder_normal
xinu@xinu-develop-end:~/raihan_2311104020$ fusermount -u ~/raihan_2311104020/folder_normal
xinu@xinu-develop-end:~/raihan_2311104020$ 
```

- e. [7 Point] Buatlah folder baru bernama folder_sembarang. Lakukan perintah berikut ini:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_sembarang
```

Amati dan tulis hasil observasi Anda!

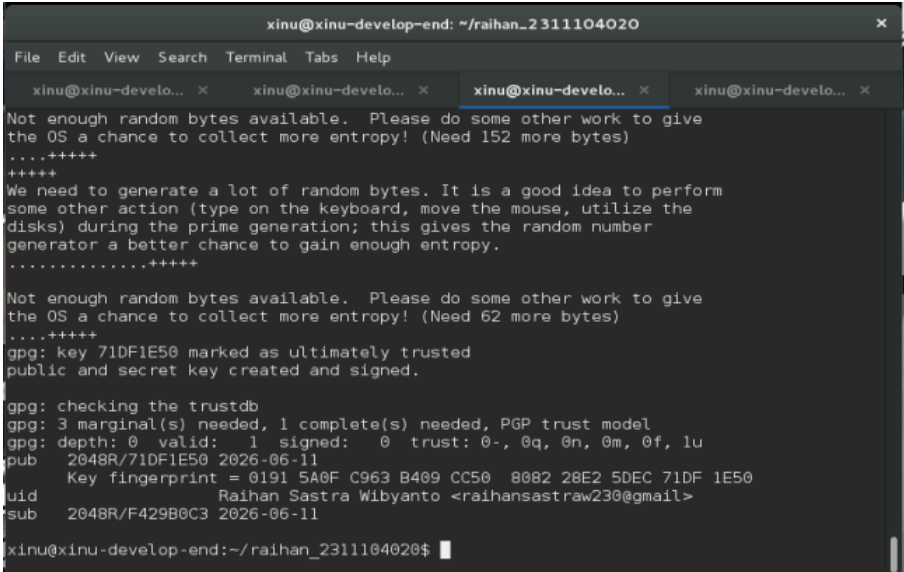
```
xinu@xinu-develop-end:~/raihan_2311104020$ encfs ~/raihan_2311104020/folder_terenkripsi
~/raihan_2311104020/folder_sembarang
EncFS Password:
xinu@xinu-develop-end:~/raihan_2311104020$
```

5. Konfidensialitas: gpg [30 Point]

- a. [5 Point] Membuat kunci publik dan privat. Jalankan perintah ini:

```
gpg --gen-key
```

Baca dan ikuti perintah yang ada dari program gpg!



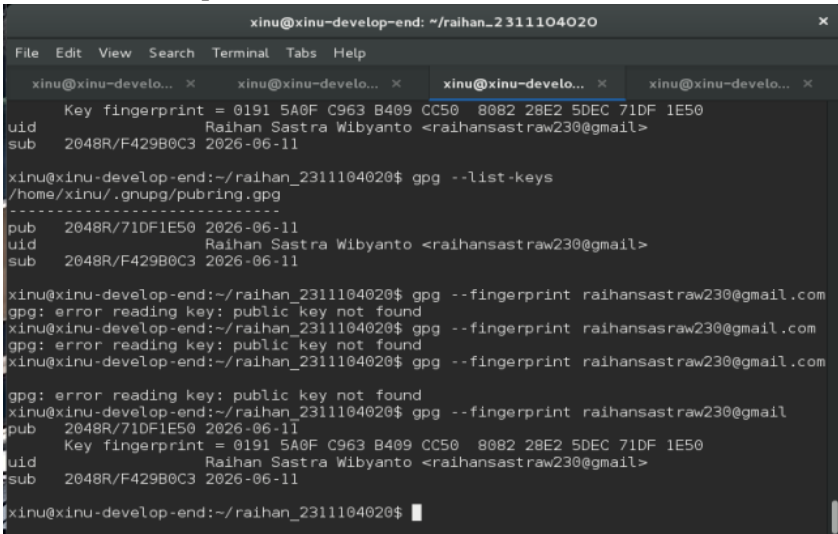
```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help
xinu@xinu-develo... xinu@xinu-develo... xinu@xinu-develo... xinu@xinu-develo...
Not enough random bytes available. Please do some other work to give
the OS a chance to collect more entropy! (Need 152 more bytes)
.....+++++
+++++
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
.....+++++
Not enough random bytes available. Please do some other work to give
the OS a chance to collect more entropy! (Need 62 more bytes)
.....+++++
gpg: key 71DF1E50 marked as ultimately trusted
public and secret key created and signed.

gpg: checking the trustdb
gpg: 3 marginal(s) needed, 1 complete(s) needed, PGP trust model
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
pub   2048R/71DF1E50 2026-06-11
      Key fingerprint = 0191 5A0F C963 B409 CC50 8082 28E2 5DEC 71DF 1E50
uid     Raihan Sastra Wibyanto <raihansastraw230@gmail>
sub     2048R/F429B0C3 2026-06-11
xinu@xinu-develop-end:~/raihan_2311104020$
```

- b. [5 Point] Hasil publickey dan privatekey ada di folder ~/.gnupg. Privatekey tidak boleh keluar dari komputer ini dan hanya Anda saja yang dapat mengaksesnya. Kunci publik akan dibagikan kepada seluruh dunia atau untuk orang yang Anda inginkan saja. Kunci publik adalah pubring.gpg dan kunci private adalah secring.gpg. Lakukan perintah berikut ini untuk mengetahui daftar key yang Anda punya dan fingerprint kunci yang Anda punya!

```
gpg --list-keys
gpg --fingerprint nama@email.com
```

Screenshot hasil perintah di atas!



```
xinu@xinu-develop-end: ~/raihan_2311104020
File Edit View Search Terminal Tabs Help
xinu@xinu-develo... xinu@xinu-develo... xinu@xinu-develo... xinu@xinu-develo...
Key fingerprint = 0191 5A0F C963 B409 CC50 8082 28E2 5DEC 71DF 1E50
uid     Raihan Sastra Wibyanto <raihansastraw230@gmail>
sub     2048R/F429B0C3 2026-06-11
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --list-keys
/home/xinu/.gnupg/pubring.gpg
pub   2048R/71DF1E50 2026-06-11
uid     Raihan Sastra Wibyanto <raihansastraw230@gmail>
sub     2048R/F429B0C3 2026-06-11
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --fingerprint raihansastraw230@gmail.com
gpg: error reading key: public key not found
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --fingerprint raihansastraw230@gmail.com
gpg: error reading key: public key not found
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --fingerprint raihansastraw230@gmail.com
gpg: error reading key: public key not found
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --fingerprint raihansastraw230@gmail
pub   2048R/71DF1E50 2026-06-11
      Key fingerprint = 0191 5A0F C963 B409 CC50 8082 28E2 5DEC 71DF 1E50
uid     Raihan Sastra Wibyanto <raihansastraw230@gmail>
sub     2048R/F429B0C3 2026-06-11
xinu@xinu-develop-end:~/raihan_2311104020$
```

- c. [5 Point] Export kunci publik Anda. Jalankan perintah ini:

```
gpg --armor --export nama@email.anda > mypublic_key.asc
```


File **mypublic_key.asc** adalah file yang akan dibagikan kepada teman-teman Anda. Teman Anda akan menggunakan kunci publik Anda jika ingin mengirim pesan rahasia kepada Anda. Hanya Anda yang dapat membaca pesan tersebut karena hanya mempunyai kunci privat yang bersesuaian dengan kunci publik Anda.

- i. Rename **mypublic_key.asc** menjadi **nim_anda.asc**, Contoh: 130118xxxx.asc
- ii. Taruh file **nim_anda.asc** ke folder pada link berikut: <http://tiny.cc/SisopNomor5C>

```
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --armor --export raihansastraw230@gmail
> mypublic_key.asc
xinu@xinu-develop-end:~/raihan_2311104020$ mv mypublic_key.asc 2311104020.asc
xinu@xinu-develop-end:~/raihan_2311104020$
```

- d. **[5 Point] Mengimport kunci publik orang lain.** Silakan download file **nim_teman_sebelah_anda.asc** dan jalankan perintah ini untuk mengimport (menambahkan kunci publik orang lain ke sistem Anda):

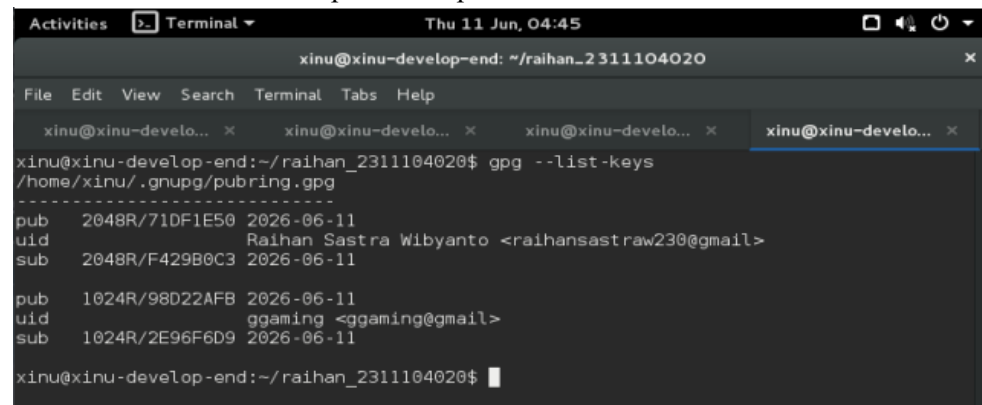
```
gpg --import nim_teman_sebelah_anda.asc
```

Contoh: “gpg --import 130118yyyy.asc” Setelah mengimport kunci publik teman Anda, Anda dapat mengirimkan pesan kepada teman Anda secara terenkripsi.

Untuk memastikan bahwa kunci publik telah diimport lakukan perintah

```
gpg --list-keys
```

dan nama teman Anda ada pada hasil perintah tersebut.



- e. **[5 Point] Enkripsi pesan.** Buatlah file bernama **file_rahasia.txt**. Isi file tersebut dengan pesan rahasia Anda. Pesan inilah yang akan Anda kirim ke teman Anda. Jalankan perintah ini untuk melakukan enkripsi:

```
//ditulis dalam satu baris
gpg --encrypt --armor -r
alamat_email_teman_anda_yang_baru_diimport@xxx.com file_rahasia.txt
xinu@xinu-develop-end:~/raihan_2311104020$ gpg --encrypt --armor -r ggaming@gmail file_
rahasia.txt
xinu@xinu-develop-end:~/raihan_2311104020$
```

- f. **[5 Point]** Hanya teman anda yang dapat membuka file tersebut. Hasil dari proses tersebut adalah **file_rahasia_untuk_teman_anda.asc** (contoh: **file_rahasia_untuk_130118yyyy.asc**) Taruh **file_rahasia_untuk_teman_anda.asc** ke folder pada link:

<http://tiny.cc/SisopNomor5F>

Download dan buka file yang diperuntukkan bagi Anda dan lakukan dekripsi untuk melihat isi pesan dengan perintah:

```
gpg file_rahasia_nim_anda.asc
```



```
xinu@xinu-develop-end:~/raihan_2311104020$ gpg -d file_rahasia.txt.asc
You need a passphrase to unlock the secret key for
user: "ggaming <ggaming@gmail>"
1024-bit RSA key, ID 2E96F6D9, created 2026-06-11 (main key ID 98D22AFB)

gpg: encrypted with 1024-bit RSA key, ID 2E96F6D9, created 2026-06-11
      "ggaming <ggaming@gmail>"
ini adalah rahasia
xinu@xinu-develop-end:~/raihan_2311104020$
```